

DATA DICTIONARY

Nova Scotia Health & Population Analytics

Star Schema | Power BI Data Model | DAX Measures Reference

Author	Team Alpha
Program	Business Intelligence & Analytics, NSCC
Version	3
Date	May 2026
Tool	Microsoft Power BI Desktop

Table of Contents

- 1. Schema Overview
- 2. Dimension Tables
 - 2.1 NovaScotia Health Zones
 - 2.2 Calendar
 - 2.3 Age Group
 - 2.4 Gender
 - 2.5 Chronic Diseases
 - 2.6 Geography
 - 2.7 Hospitalization Diagnosis
 - 2.8 Dim_Correlation_Matrix
- 3. Fact Tables
 - 3.1 Population
 - 3.2 Life Expectancy
 - 3.3 All Chronic Diseases

- 3.4 Health Zone Population
- 3.5 Hospitalization Causes
- 3.6 Hospitalization Numbers
- 3.7 chronic_disease_report

- 4. Relationships
- 5. DAX Measures Reference

1. Schema Overview

This data model follows a star schema design optimized for Power BI analytics. It consists of 8 dimension tables and 7 fact tables covering Nova Scotia population demographics, chronic disease prevalence, hospitalization patterns, surgical volumes, alternate level of care (ALC) metrics, socioeconomic-health correlation analysis, and chronic disease population projections.

The model supports analysis across four NS health zones, national benchmarking against all Canadian provinces, time-series trending from 2000–2035 (calendar) with fact data spanning 2012 - 2025, and demographic segmentation by age group and gender.

Model summary

Component	Count	Details
Dimension Tables	8	Health Zone, Calendar, Age Group, Gender, Chronic Disease, Geography, Diagnosis, Correlation Matrix
Fact Tables	7	Population, Life Expectancy, Chronic Disease, Health Zone Population, Hospitalization Causes, Hospitalization Numbers, Chronic Disease Report
DAX Measures	21	Population (7), Chronic Disease (4), Health Zone (5), Hospitalization (4), ALC (1), Correlation Matrix (5)
Data Span	2012–2025	Calendar dimension extends 2000–2035 for projection support

2. Dimension Tables

2.1 NovaScotia Health Zones

Defines the 4 Nova Scotia Health Authority management zones. Contains spatial geometry (MULTIPOLYGON) for Power BI shape map rendering and area calculations for population density.

Column	Data Type	Key Type	Description
Health ZoneID	Integer	Primary Key	Unique zone identifier (1–4)
Health Zone	Text	—	Zone code (ZONE 1, ZONE 2, etc.)
Health Zone Name	Text	—	Descriptive name (Western, Northern, Eastern, Central)
Geom	Text (WKT)	—	MULTIPOLYGON geometry for map visuals
Shape Length	Decimal	—	Perimeter in coordinate units
Shape Area	Decimal	—	Area in coordinate units; used in Population Density measure

Row count: 4 | Grain: One row per health zone

2.2 Calendar

Comprehensive date dimension spanning 2000–2035+ with fiscal year support (April 1 start), NS holiday flags, flu season indicator, ISO week numbering, and bi-weekly pay period tracking.

Column	Data Type	Key Type	Description
DateID	Date	Primary Key	Calendar date (M/D/YYYY format)
Year	Integer	—	Calendar year (2000–2035+)
Month / Month Name	Int / Text	—	Month number (1–12) and name
Fiscal Year / FQ / FM	Int / Text	—	NS fiscal year (Apr 1 start), fiscal quarter (FQ1–FQ4), fiscal month (1–12)
Is Holiday / Holiday Name	Boolean / Text	—	NS statutory holidays (New Year's, Good Friday, Canada Day, etc.)
Is Flu Season	Boolean	—	TRUE for Nov–Mar; supports seasonal hospitalization analysis
Is Working Day	Boolean	—	TRUE for non-weekend, non-holiday days
ISO Week / ISO Year	Integer	—	ISO 8601 week numbering
Pay Period Start/End/#	Date / Int	—	Bi-weekly pay period boundaries

Row count: ~13,000+ | Grain: One row per calendar day

2.3 Age Group

Multi-granularity age classification supporting standard 5-year census bands, CCDSS wider bands (35–44, 45–54, etc.), and broad analytical bands (Children, Youth, Working Age, Senior). Enables flexible age aggregation across different fact tables.

Column	Data Type	Key Type	Description
AgeGroupKey	Integer	Primary Key	Unique key (1000–1020 for 5-yr bands; 3000–3004 for CCDSS; 4000 for 20–39)
AgeGroupLabel	Text	—	Display label (e.g. “0 to 4 years”, “35 to 44”)
CommonBand	Text	—	Decade grouping (0 to 9, 10 to 19, etc.)
AgeBandMin / AgeBandMax	Integer	—	Numeric range boundaries for calculations
BandSortOrder	Integer	—	Sort key for visual ordering (1–9)
BroadBand	Text	—	Analytical grouping: Children, Youth, Working Age, Senior

Row count: ~26 | Grain: One row per age band definition

2.4 Gender

Simple gender dimension with display-friendly labels. The “All” row (ID 3000) supports aggregated views where gender breakdown is not applicable.

Column	Data Type	Key Type	Description
Gender ID	Integer	Primary Key	1000=F, 2000=M, 3000=All
Gender	Text	—	Short code (F, M, All)

Gender+	Text	—	Display label (Women, Men, Not Defined)
---------	------	---	---

Row count: 3 | Grain: One row per gender category

2.5 Chronic Diseases

Defines the 6 chronic diseases tracked through the Canadian Chronic Disease Surveillance System (CCDSS) data. Covers cardiovascular, respiratory, and metabolic conditions.

Column	Data Type	Key Type	Description
Chronic Disease ID	Integer	Primary Key	Unique ID (1–6)
Chronic Disease	Text	—	Disease name (IHD, AMI Crude, Asthma, COPD, Diabetes, Hypertension)

Row count: 6 | Grain: One row per chronic disease

2.6 Geography

National geography dimension covering all provinces, territories, and Canada-level aggregates. Includes regional grouping and Boolean flags for NS and national filtering, enabling provincial benchmarking.

Column	Data Type	Key Type	Description
GeoKey	Integer	Primary Key	Unique geography ID (1–15); NS = 5
ProvinceName	Text	—	Full name (e.g. Nova Scotia, Ontario)
ProvinceAbbrev / Short	Text	—	Standard abbreviation (N.S.) and short code (NS)
Region	Text	—	Regional grouping: Atlantic, Central, Prairies, Pacific, Territories, National
GeoLevel	Text	—	National, Province, or Territory
IsNS / IsNational	Boolean	—	Filter flags for NS-specific and Canada-level analysis
SortOrder	Integer	—	Display sort (1=Canada, 5=NS, etc.)

Row count: 15 | Grain: One row per province/territory/national aggregate

2.7 Hospitalization Diagnosis

Lists the most responsible diagnoses for inpatient hospitalizations as reported by CIHI. Covers obstetric, respiratory, cardiovascular, mental health, musculoskeletal, digestive, and other conditions.

Column	Data Type	Key Type	Description
Diagnosis ID	Integer	Primary Key	Unique ID (10000–10036)
Most responsible diagnosis...	Text	—	Full diagnosis description

Row count: 37 | Grain: One row per diagnosis category

2.8 Dim_Correlation_Matrix

Pre-computed Pearson correlation results between socioeconomic variables (Median Income, Average Income, Tertiary Education %) and health outcomes (Life Expectancy, Chronic Disease Burden, and specific chronic diseases) across Nova Scotia health zones. Supports the socioeconomic-health equity analysis visuals.

Column	Data Type	Key Type	Description
Socioeconomic_Variable	Text	—	Internal variable name for the socioeconomic indicator (e.g.

			Median_Income, Average_Income, Tertiary_Edu_Pct)
Socioeconomic_Label	Text	—	Display-friendly label for the socioeconomic variable (e.g. Median Income, Average Income, Tertiary Edu %)
Health_Variable	Text	—	Internal variable name for the health outcome (e.g. Life_Expectancy, Chronic_Disease_Burden, Acute Myocardial Infarction (AMI) Crude)
Health_Label	Text	—	Display-friendly label for the health outcome (e.g. Life Expectancy, Disease Burden (Avg), AMI)
Pearson_r	Decimal	Measure	Pearson correlation coefficient (−1 to +1). Positive values indicate higher socioeconomic status correlates with better health outcomes; negative values indicate inverse relationship.
Abs_r	Decimal	Measure	Absolute value of Pearson_r. Used for sorting and visual encoding by strength regardless of direction.
P_Value	Decimal	Measure	Two-tailed p-value for the correlation. Values below 0.05 are statistically significant (see Significant column).
Significant	Boolean	—	TRUE if P_Value < 0.05; FALSE otherwise. Used to highlight statistically significant correlations in visuals.
Strength	Text	—	Categorical label derived from Abs_r: Weak (<0.3), Moderate (0.3–0.59), Strong (0.6–0.79), Very Strong (≥0.8).
Direction	Text	—	Interpretive label based on sign of Pearson_r and the health variable: Risk (higher SES, higher disease), Protective (higher SES, lower disease), Beneficial (higher SES, better outcome), Harmful, or Detection Effect (where higher income enables earlier diagnosis, inflating apparent prevalence).

Row count: 40 | Grain: One row per socioeconomic variable × health variable combination

Additional columns and measures in dim_correlation_matrix:

Name	Type	Data Type	Description
Pair_Label	Calc. Column	Text	Concatenated label for the socioeconomic–health variable pair (e.g. “Median Income Life Expectancy”). Used as an axis or tooltip label in correlation visuals.
Lollipop Color	Calc. Column	Text (hex colour)	Returns a hex colour string via SWITCH() on Direction: Protective=#185FA5, Risk=#A32D2D, Detection Effect=#D85A30, Beneficial=#0F6E56, Harmful=#7B2D8B, default=#888780. Drives conditional colour on lollipop chart visuals.
Correlation Score	Measure	Decimal (constant)	Hardcoded reference constant = 0.91. Used as a positive-threshold annotation line on the correlation chart.

Correlation score2	Measure	Decimal (constant)	Hardcoded reference constant = -0.81. Paired with Correlation Score to annotate the negative-end reference line on the correlation chart.
Pearson R	Measure	Decimal	Returns the Pearson_r value for the currently selected row via <code>SELECTEDVALUE(dim_correlation_matrix[Pearson_r])</code> . Used in card visuals and tooltips to show the correlation coefficient for the user-selected variable pair.
Thank you	Measure	Text (constant)	Returns the static string "Thank You –Team Alpha". Used as a closing caption on the final report page.

3. Fact Tables

3.1 Population

Annual population estimates by gender, age group, and geography for Nova Scotia and all Canadian provinces. Sources from Statistics Canada annual demographic estimates (2012–2025). Enables NS vs national benchmarking and demographic trend analysis.

Column	Data Type	Key Type	Description
Year	Integer	Degenerate dim	Calendar year (2012–2025)
Gender ID	Integer	FK → DimGender	1000=F, 2000=M
AgeGroupKey	Integer	FK → DimAgeGroup	5-year age band key
GeoKey	Integer	FK → DimGeography	Province/territory (5=NS, 1=Canada)
Population	Integer	Measure	Population count for the year/gender/age/geo combination

Grain: One row per year × gender × age group × geography

3.2 Life Expectancy

Canadian male life expectancy by single year of age and 3-year rolling period. Used for life expectancy trend analysis and health outcome benchmarking.

Column	Data Type	Key Type	Description
Gender ID	Integer	FK → DimGender	Currently male only (2000)
Age	Text	—	Single year of age ("0 years" to "76 years"+)
Year Period	Text	—	3-year rolling window (e.g. "2022 to 2024")
Life Expectancy	Decimal	Measure	Expected remaining years of life at given age

Grain: One row per gender × age × 3-year period

3.3 All Chronic Diseases

Chronic disease prevalence counts by health zone, gender, age group, and year. Two grain levels exist: 5-year age bands (AgeGroupKey 1000–1020) for IHD and similar conditions, and CCDSS wider bands (3000–3004) for COPD. Also includes AMI young-adult data (AgeGroupKey 4000, age 20–39). Source: CCDSS via Public Health Agency of Canada.

Column	Data Type	Key Type	Description
Date	Date	FK → DimDate	January 1 of reporting year

Gender ID	Integer	FK → DimGender	1000=F, 2000=M, 3000=All (for AMI young adults)
Chronic Disease ID	Integer	FK → DimChronicDisease	Disease identifier (1–6)
Health ZoneID	Integer	FK → DimHealthZone	NS health zone (1–4)
AgeGroupKey	Integer	FK → DimAgeGroup	Age band key (multiple grains)
prevalence	Integer	Measure	Count of individuals with the condition
population	Integer	Measure	Denominator population for rate calculation

Grain: One row per date × gender × disease × zone × age group

3.4 Health Zone Population

2021 Census population by health zone, age group, and gender. Used for zone-level demographic profiling, population weights, population pyramids, and population density calculations.

Column	Data Type	Key Type	Description
Census Year	Integer	—	2021
Health ZoneID	Integer	FK → DimHealthZone	NS health zone (1–4)
AgeGroupKey	Integer	FK → DimAgeGroup	5-year age band
Gender ID	Integer	FK → DimGender	1000=F, 2000=M
Population	Integer	Measure	Population count

Grain: One row per zone × age group × gender (2021 census only)

3.5 Hospitalization Causes

Inpatient hospitalization volumes and average length of stay by most responsible diagnosis, geography, and fiscal year. Source: CIHI Discharge Abstract Database. Enables NS vs national benchmarking by condition.

Column	Data Type	Key Type	Description
DateID	Date	FK → DimDate	Fiscal year start date (April 1)
GeoKey	Integer	FK → DimGeography	Province/territory/national
Diagnosis ID	Integer	FK → DimDiagnosis	Most responsible diagnosis
Fiscal Year	Integer	—	Fiscal year label (e.g. 2018, 2019)
Number of inpatient hospitalizations	Integer	Measure	Count of inpatient stays
Percentage of inpatient hospitalizations	Decimal	Measure	Share of total hospitalizations (%)
Average acute length of stay	Decimal	Measure	Average days per inpatient stay

Grain: One row per fiscal year × geography × diagnosis

3.6 Hospitalization Numbers

Aggregate hospitalization volumes with Alternate Level of Care (ALC) metrics. Compares NS to Canada (excl. Quebec) across fiscal years. Key table for ALC bed-blocking analysis.

Column	Data Type	Key Type	Description
DateID	Date	FK → DimDate	Fiscal year start (April 1)
GeoKey	Integer	FK → DimGeography	2=Canada excl. QC, 5=NS

Number of hospitalizations	Integer	Measure	Total inpatient hospitalizations
...with reported ALC days	Integer	Measure	Hospitalizations involving ALC designation
Proportion...ALC days (%)	Decimal	Measure	% of hospitalizations with ALC
Total LOS (days)	Integer	Measure	Total length of stay in days
ALC LOS (days)	Integer	Measure	Days designated as ALC
Patient days in ALC (%)	Decimal	Measure	ALC days as % of total LOS

Grain: One row per fiscal year × geography

3.7 chronic_disease_report

Chronic disease population and projection data by health zone, gender, age group, chronic disease, and date. Extends the All Chronic Diseases fact table with predicted population and predicted prevalence columns to support forecasting and trend analysis on the chronic disease report pages.

Column	Data Type	Key Type	Description
AgeGroupKey	Integer	FK → DimAgeGroup	Age band key linking to DimAgeGroup.
Chronic Disease ID	Integer	FK → DimChronicDisease	Disease identifier (1–6) linking to DimChronicDisease.
Date	Date	FK → DimDate	January 1 of reporting year; links to DimDate.
Gender ID	Integer	FK → DimGender	1000=F, 2000=M, 3000=All.
Health ZoneID	Integer	FK → DimHealthZone	NS health zone identifier (1–4).
population	Integer	Measure	Denominator population for the zone/age/gender/disease/year combination.
predicted_population	Decimal	Measure	Projected future population for the same zone/age/gender/disease/year slice. Used to model how population size is expected to change for forecasting purposes.
predicted_prevalence	Decimal	Measure	Projected number of individuals expected to have the condition in a future year. Enables forward-looking chronic disease burden analysis.

Grain: One row per date × gender × chronic disease × health zone × age group

4. Relationships

All relationships are single-direction (fact → dimension), many-to-one cardinality, and active unless noted. The star schema ensures each fact table connects to shared dimensions for consistent cross-filtering.

From (Fact/Dim)	To (Dimension)	Cardinality	Status
FactPopulation[Gender ID]	DimGender[Gender ID]	Many:1	Active
FactPopulation[AgeGroupKey]	DimAgeGroup[AgeGroupKey]	Many:1	Active
FactPopulation[GeoKey]	DimGeography[GeoKey]	Many:1	Active
FactChronicDisease[Date]	DimDate[DateID]	Many:1	Active
FactChronicDisease[Gender ID]	DimGender[Gender ID]	Many:1	Active

FactChronicDisease[Chronic Disease ID]	DimChronicDisease[Chronic Disease ID]	Many:1	Active
FactChronicDisease[Health ZoneID]	DimHealthZone[Health ZoneID]	Many:1	Active
FactChronicDisease[AgeGroupKey]	DimAgeGroup[AgeGroupKey]	Many:1	Active
FactHealthZonePopulation[Health ZoneID]	DimHealthZone[Health ZoneID]	Many:1	Active
FactHealthZonePopulation[AgeGroupKey]	DimAgeGroup[AgeGroupKey]	Many:1	Active
FactHealthZonePopulation[Gender ID]	DimGender[Gender ID]	Many:1	Active
FactHospitalization[DateID]	DimDate[DateID]	Many:1	Active
FactHospitalization[GeoKey]	DimGeography[GeoKey]	Many:1	Active
FactHospitalization[Diagnosis ID]	DimDiagnosis[Diagnosis ID]	Many:1	Active
FactHospitalizationNumbers[DateID]	DimDate[DateID]	Many:1	Active
FactHospitalizationNumbers[GeoKey]	DimGeography[GeoKey]	Many:1	Active

5. DAX Measures Reference

All measures organized by source table. Each measure includes its DAX formula and a plain-language description of what it calculates and how it should be used.

All Chronic Diseases (FactChronicDisease)

Total Prevalence	SUM('All Chronic Diseases'[prevalence])	Sums the raw count of individuals diagnosed with a chronic condition across the current filter context (disease, zone, gender, age, year). Base building block for prevalence rate calculations.
Chronic Population Denominator	SUM('All Chronic Diseases'[population])	Sums the reference population used as the denominator when calculating prevalence rates. This population is specific to the disease/zone/age/gender/year combination and comes directly from CCDSS data.
Prevalence Rate per 1000	DIVIDE([Total Prevalence], [Chronic Population Denominator]) * 1000	Calculates the number of people with a chronic condition per 1,000 population. Enables fair comparison across zones and demographics with different population sizes. Example: a rate of 85 means 85 people per 1,000 have the condition.
Zone Disparity Ratio	DIVIDE(MAXX(ALL('NovaScotia Health Zones'[Health Zone]), [Prevalence Rate per 1000]), MINX(ALL('NovaScotia Health Zones'[Health Zone]), [Prevalence Rate per 1000]))	Divides the highest zone prevalence rate by the lowest zone prevalence rate, ignoring any zone filter. A ratio of 1.47 means the worst zone has 47% higher disease burden than the best. Useful for identifying geographic health inequities.

Health Zone Population (FactHealthZonePopulation)

Population(total)	SUM('Health Zone Population'[Population])	Total zone population from the 2021 Census. Base measure for zone-level demographics. Responds to DimHealthZone, DimAgeGroup, and DimGender slicers via foreign key relationships.
Male Population	CALCULATE([Population(total)], Gender[Gender+] = "Men")	Filters zone population to men only. Used on the right side of the population pyramid visual.
Female Population	CALCULATE([Population(total)], Gender[Gender+] = "Women") * -1	Filters zone population to women and multiplies by -1 to create the left-side (negative) axis of the population pyramid. The negative value makes the pyramid mirror correctly.
Zone Weight %	DIVIDE([Population(total)], CALCULATE([Population(total)], ALL('NovaScotia Health Zones')))	Each zone's share of the total NS population. ALL() removes zone filters so the denominator is always the full NS total. Used to show that Zone 4 (Central/HRM) holds ~41% of the province's population.
Population Density	DIVIDE([Population(total)], MAX('NovaScotia Health Zones'[Shape Area]) / 1000000)	Population per square unit of area. Divides zone population by the geographic area from DimHealthZone (converted from coordinate units to a readable scale). Highlights that Central is densely populated while Western/Eastern are sparse.

Population (FactPopulation)

Total Population	SUM(Population[Population])	Grand total population across all genders and age groups for the current geography and year context. Foundation measure for all population analytics.
NS Population	CALCULATE([Total Population], Geography[ProvinceName] = "Nova Scotia")	Forces the geography filter to Nova Scotia regardless of slicer context. Ensures NS-specific KPI cards always show the correct value.
NS Population (2025)	CALCULATE(SUM(Population[Population]), Geography[ProvinceName] = "Nova Scotia", 'Calendar'[Year] = 2025)	Hardcoded to return the 2025 NS population estimate. Used in headline KPI cards where you want a fixed reference point rather than a dynamic slicer-driven value.
NS Population Growth Since 2012	VAR Pop2025 = CALCULATE(..., Year=2025) VAR Pop2012 = CALCULATE(..., Year=2012) VAR GrowthPercent = DIVIDE(Pop2025-Pop2012, Pop2012) RETURN "+" & FORMAT(GrowthPercent, "0.0%") & " since 2012"	Calculates the total percentage growth between 2012 and 2025 and returns a formatted text string (e.g. "+12.5% since 2012"). Used as a subtitle in KPI cards to provide historical context.

Senior Population	<code>CALCULATE([Total Population], 'Age Group'[BroadBand] = "Senior")</code>	Filters to the Senior broad band (ages 65+) from DimAgeGroup. Used in dependency ratio calculations and to track the aging population trend.
Working Age Population	<code>CALCULATE([Total Population], 'Age Group'[BroadBand] = "Working Age")</code>	Filters to Working Age broad band (ages 20–64). The denominator in dependency ratio calculations and a key indicator of economic capacity and tax base.

Hospitalization Causes (FactHospitalization)

Total inpatient Hospitalizations	<code>SUM('Hospitalization Causes' [Number of inpatient hospitalizations])</code>	Total inpatient hospitalization count from the diagnosis-level fact table. Sliceable by geography, diagnosis, and fiscal year. Used for NS vs national diagnosis comparisons.
---	---	---

Hospitalization Numbers (FactHospitalizationNumbers)

Total Hospitalizations	<code>SUM('Hospitalization Numbers' [Number of hospitalizations])</code>	Aggregate hospitalization count from the ALC fact table. Unlike the diagnosis-level measure, this is the province-wide total used for ALC rate calculations.
ALC Hospitalization Rate %	<code>DIVIDE(SUM('Hospitalization Numbers' [...with reported ALC days]), SUM('Hospitalization Numbers' [Number of hospitalizations]))</code>	Percentage of all hospitalizations that involve at least one ALC-designated day. Rising ALC rates indicate growing bed-blocking pressure where patients are medically ready for discharge but have nowhere to go (e.g. awaiting long-term care placement).
Hospitalization Rate per 1000	<code>DIVIDE([Total Hospitalizations], [Total Population]) * 1000</code>	Hospitalizations per 1,000 population. Standardizes volumes against population size for fair geographic comparison. Requires both FactHospitalizationNumbers and FactPopulation to be in context.